OTHERS HELP

# VizieR

**Search Criteria**
Save in CDSportal
**Keywords** Back
- V/154/sdss16

**Tables** Add
V/154
..sdss16
Choose

**Preferences**
max: 999999
| -Separated-Values
All columns
Compute
Distance {rho}
Position angle {theta}
Distance (x,y)
Galactic
J2000
B1950
Ecl. J2000
default
Sort by Distance
+ order - 
No sort

**Position** in:
Sexagesimal
Decimal °
Truncated prec.

**Time**
Time JD
Time ISO

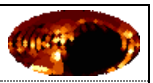
**Mirrors**
CDS, France

Simple Target | List Of Targets — Fast Xmatch with large catalogs or Simbad

**Target** Name (resolved by Sesame) or **Position**: Target dimension:
Clear J2000 2 arcmin

*NB: The epoch used for the query is the original epoch of the table(s)* Radius Box size

**V/154** Sloan Digital Sky Surveys (SDSS), Release 16 (DR16) (Ahumada+, 2020)
Similar Catalogs 2020ApJS..249....3A ReadMe+ftp
Post annotation



1.V/154/sdss16 Data Release 16 (469,050,976 *primary* sources plus 325,277,739 *secondary* sources) (original column names in green) (Note) *(1231051050 rows)*

Simple Constraint | List Of Constraints — Submit Reset All

**Query by Constraints applied on Columns** *(Output Order:* + - *)*
Standard Original

| *Show* | *Sort* | *Column* | Clear *Constraint* | | *Explain* *(UCD)* |
|---|---|---|---|---|---|
| ☑ | ○ | objID | | | (i) SDSS unique object identifier (1) (links to online SDSS-DR16 data) (Note) (meta.id;meta.main) |
| ☑ | ○ | RA_ICRS | | deg | (i) Right Ascension of the object (ICRS) at Epoch="MJD" (ra) (pos.eq.ra;meta.main) |
| ☑ | ○ | DE_ICRS | | deg | (i) Declination of the object (ICRS) at Epoch="MJD" (dec) (pos.eq.dec;meta.main) |
| ☑ | ○ | fieldID | | | The field this object is in (meta.note) |
| ☑ | ○ | mode | =1 | | [1/4] PhotoMode (1=primary, 2=secondary, 3=family, 4=outside) (qualified to =1 by default) (meta.code.class) |
| ☑ | ○ | class | =3 | | (n) [0/8] Type of object (3=galaxy, 6=star) (type) (Note 2) (src.class) |
| ☑ | ○ | clean | =1 | | [0/1] Clean photometry flag (1=clean; 0=unclean) (qualified to =1 by default) (meta.code) |
| ☑ | ○ | flags | | | Photo Object Attribute flags (Note 3) (meta.code.error) |
| ☑ | ○ | upmag | | mag | (n)(i) [4/35] PSF magnitude in u band (psfMag_u) (Note 4) (phot.mag;em.opt.U) |
| ☑ | ○ | gpmag | | mag | (n) [3.7/37] PSF magnitude in g band (psfMag_g) (Note 4) (phot.mag;em.opt.B) |
| ☑ | ○ | rpmag | | mag | (n) [3.8/36] PSF magnitude in r band (psfMag_r) (Note 4) (phot.mag;em.opt.R) |
| ☑ ALL cols | Reset All | | Clear | | (n) indicates a possible blank or NULL column (i) indexed column Submit |
| ☑ | ○ | ipmag | | mag | (n) [2.6/34] PSF magnitude in i band (psfMag_i) (Note 4) (phot.mag;em.opt.I) |
| ☑ | ○ | zpmag | | mag | (n) [3.8/33.3] PSF magnitude in z band (psfMag_z) (Note 4) (phot.mag;em.opt.I) |
| ☑ | ○ | e_upmag | | mag | (n) [1.5e-5/100] Mean error on upmag (psfMagErr_u) (Note 4) (stat.error;phot.mag) |
| ☑ | ○ | e_gpmag | | mag | (n) Mean error on gpmag (psfMagErr_g) (Note 4) (stat.error;phot.mag) |
| ☑ | ○ | e_rpmag | | mag | (n) Mean error on rpmag (psfMagErr_r) (Note 4) (stat.error;phot.mag) |
| ☑ | ○ | e_ipmag | | mag | (n) Mean error on ipmag (psfMagErr_i) (Note 4) (stat.error;phot.mag) |
| ☑ | ○ | e_zpmag | | mag | (n) Mean error on zpmag (psfMagErr_z) (Note 4) (stat.error;phot.mag) |
| ☑ | ○ | uPmag | | mag | (n) [-0.4/47.2] Petrosian magnitude in u band (petroMag_u) (Note 4) (phot.mag;em.opt.U) |
| ☑ | ○ | gPmag | | mag | (n) [3.7/48] Petrosian magnitude in g band (petroMag_g) (Note 4) (phot.mag;em.opt.B) |

| | | | | |
|---|---|---|---|---|
| ☑ | ○ | rPmag | mag | (n) [1.3/48.6] Petrosian magnitude in r band (petroMag_r) *(Note 4)* (phot.mag;em.opt.R) |
| ☑ | ○ | iPmag | mag | (n) [2.6/48.4] Petrosian magnitude in i band (petroMag_i) *(Note 4)* (phot.mag;em.opt.I) |
| ☑ | ○ | zPmag | mag | (n) [-2.9/45.1] Petrosian magnitude in z band (petroMag_z) *(Note 4)* (phot.mag;em.opt.I) |
| ☑ | ○ | e_uPmag | mag | (n) [1e-7/100] Mean error in uPmag (petroMagErr_u) *(Note 4)* (stat.error;phot.mag) |

☑ **ALL** cols Reset All Clear (n) indicates a possible blank or NULL column (i) indexed column Submit

| | | | | |
|---|---|---|---|---|
| ☑ | ○ | e_gPmag | mag | (n) [9.1e-9/100] Mean error in gPmag (petroMagErr_g) *(Note 4)* (stat.error;phot.mag) |
| ☑ | ○ | e_rPmag | mag | (n) Mean error in rPmag (petroMagErr_r) *(Note 4)* (stat.error;phot.mag) |
| ☑ | ○ | e_iPmag | mag | (n) Mean error in iPmag (petroMagErr_i) *(Note 4)* (stat.error;phot.mag) |
| ☑ | ○ | e_zPmag | mag | (n) Mean error in zPmag (petroMagErr_z) *(Note 4)* (stat.error;phot.mag) |
| ☑ | ○ | uPrad | arcsec | (n) [1.2e-6/261] Petrosian radius in u band (petroRad_u) (phys.angSize;src) |
| ☑ | ○ | gPrad | arcsec | (n) [2.3e-6/261] Petrosian radius in g band (petroRad_g) (phys.angSize;src) |
| ☑ | ○ | rPrad | arcsec | (n) [4.9e-6/260] Petrosian radius in r band (petroRad_r) (phys.angSize;src) |
| ☑ | ○ | iPrad | arcsec | (n) [2.2e-6/263] Petrosian radius in i band (petroRad_i) (phys.angSize;src) |
| ☑ | ○ | zPrad | arcsec | (n) [8.3e-7/261] Petrosian radius in z band (petroRad_z) (phys.angSize;src) |
| ☑ | ○ | e_uPrad | arcsec | (n) [-1000/259] Mean error in rPrad (petroRadErr_u) (stat.error) |
| ☑ | ○ | e_gPrad | arcsec | (n) [-1000/259] Mean error in rPrad (petroRadErr_g) (stat.error) |
| ☑ | ○ | e_rPrad | arcsec | (n) [-1000/259] Mean error in rPrad (petroRadErr_r) (stat.error) |
| ☑ | ○ | e_iPrad | arcsec | (n) [-1000/259] Mean error in rPrad (petroRadErr_i) (stat.error) |

☑ **ALL** cols Reset All Clear (n) indicates a possible blank or NULL column (i) indexed column Submit

| | | | | |
|---|---|---|---|---|
| ☑ | ○ | e_zPrad | arcsec | (n) [-1000/257] Mean error in rPrad (petroRadErr_z) (stat.error) |
| ☑ | ○ | udVrad | arcsec | (n) [0/257] De Vaucouleurs fit radius u band (deVRad_u) (phys.angSize;src) |
| ☑ | ○ | gdVrad | arcsec | (n) [0/238] De Vaucouleurs fit radius g band (deVRad_g) (phys.angSize;src) |
| ☑ | ○ | rdVrad | arcsec | (n) [0/238] De Vaucouleurs fit radius r band (deVRad_r) (phys.angSize;src) |
| ☑ | ○ | idVrad | arcsec | (n) [0/232] De Vaucouleurs fit radius i band (deVRad_i) (phys.angSize;src) |
| ☑ | ○ | zdVrad | arcsec | (n) [0/238] De Vaucouleurs fit radius z band (deVRad_z) (phys.angSize;src) |
| ☑ | ○ | udVell | | (n) [0.05/1] De Vaucouleurs fit b/a u band (deVAB_u) (phys.size.axisRatio) |
| ☑ | ○ | gdVell | | (n) [0.05/1] De Vaucouleurs fit b/a g band (deVAB_g) (phys.size.axisRatio) |
| ☑ | ○ | rdVell | | (n) [0.05/1] De Vaucouleurs fit b/a r band (deVAB_r) (phys.size.axisRatio) |
| ☑ | ○ | idVell | | (n) [0.05/1] De Vaucouleurs fit b/a i band (deVAB_i) (phys.size.axisRatio) |
| ☑ | ○ | zdVell | | (n) [0.05/1] De Vaucouleurs fit b/a z band (deVAB_z) (phys.size.axisRatio) |
| ☑ | ○ | uPA | deg | (n) [-10105/271] Position angle of dVrad in u band (deVPhi_u) (pos.posAng) |
| ☑ | ○ | gPA | deg | (n) [-10105/271] Position angle of dVrad in g band (deVPhi_g) (pos.posAng) |

ALL cols | Reset All | Clear | (n) indicates a possible blank or NULL column (i) indexed column | Submit

| | | | | |
|---|---|---|---|---|
| rPA | | deg | (n) [-10105/271] Position angle of dVrad in r band (deVPhi_r) (pos.posAng) |
| iPA | | deg | (n) [-10105/271] Position angle of dVrad in i band (deVPhi_i) (pos.posAng) |
| zPA | | deg | (n) [-10105/271] Position angle of dVrad in z band (deVPhi_z) (pos.posAng) |
| uFlags | | | Detection flags u band (flags_u) *(Note 3)* (meta.code) |
| gFlags | | | Detection flags g band (flags_g) *(Note 3)* (meta.code) |
| rFlags | | | Detection flags r band (flags_r) *(Note 3)* (meta.code) |
| iFlags | | | Detection flags i band (flags_i) *(Note 3)* (meta.code) |
| zFlags | | | Detection flags z band (flags_z) *(Note 3)* (meta.code) |
| uc | | | (n) [0/8] Phototype in u band (6=Star ; 3=galaxy) (type_u) *(Note 2)* (src.class.starGalaxy) |
| gc | | | (n) [0/8] Phototype in i band (6=Star ; 3=galaxy) (type_g) *(Note 2)* (src.class.starGalaxy) |
| rc | | | (n) [0/8] Phototype in r band (6=Star ; 3=galaxy) (type_r) *(Note 2)* (src.class.starGalaxy) |
| ic | | | (n) [0/8] Phototype in i band (6=Star ; 3=galaxy) (type_i) *(Note 2)* (src.class.starGalaxy) |
| zc | | | (n) [0/8] Phototype in z band (6=Star ; 3=galaxy) (type_z) *(Note 2)* (src.class.starGalaxy) |

ALL cols | Reset All | Clear | (n) indicates a possible blank or NULL column (i) indexed column | Submit

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|---|---|---|---|---|
| us | | | | (n) [0/1] 0=notStar 1=Star in u band (probPSF_u) (src.class.starGalaxy) |
| gs | | | | (n) [0/1] 0=notStar 1=Star in g band (probPSF_g) (src.class.starGalaxy) |
| rs | | | | (n) [0/1] 0=notStar 1=Star in r band (probPSF_r) (src.class.starGalaxy) |
| is | | | | (n) [0/1] 0=notStar 1=Star in i band (probPSF_i) (src.class.starGalaxy) |
| zs | | | | (n) [0/1] 0=notStar 1=Star in z band (probPSF_z) (src.class.starGalaxy) |
| e_RA_ICRS | | arcsec | | [0/51703] Mean error on RAdeg (raErr) (stat.error;pos.eq.ra) |
| e_DE_ICRS | | arcsec | | [0/22795] Mean error on DEdeg (decErr) (stat.error;pos.eq.dec) |
| uoRA | | arcsec | | Offset of u position along RA (offsetRa_u) (pos.angDistance) |
| goRA | | arcsec | | Offset of g position along RA (offsetRa_g) (pos.angDistance) |
| roRA | | arcsec | | Offset of r position along RA (offsetRa_r) (pos.angDistance) |
| ioRA | | arcsec | | Offset of i position along RA (offsetRa_i) (pos.angDistance) |
| zoRA | | arcsec | | Offset of z position along RA (offsetRa_z) (pos.angDistance) |
| uoDE | | arcsec | | Offset of u position along Dec (offsetDec_u) (pos.angDistance) |

ALL cols | Reset All | Clear | (n) indicates a possible blank or NULL column (i) indexed column | Submit

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|---|---|---|---|---|
| goDE | | arcsec | | Offset of g position along Dec (offsetDec_g) (pos.angDistance) |
| roDE | | arcsec | | Offset of r position along Dec (offsetDec_r) (pos.angDistance) |
| ioDE | | arcsec | | Offset of i position along Dec (offsetDec_i) (pos.angDistance) |
| zoDE | | arcsec | | Offset of z position along Dec (offsetDec_z) (pos.angDistance) |
| parentID | | | | Pointer to parent (if object deblended) (meta.id.parent) |
| SpObjID | | | (char) | (i) Pointer to the spectrum of object or 0 (specObjID) *(Note 6)* (meta.id) |
| umag | | mag | | (n)(i) [4/37.3] Model magnitude in u filter AB scale (u) *(Note 7)* (phot.mag;em.opt.U) |
| gmag | | mag | | (n)(i) [3.7/39.6] Model magnitude in g filter AB scale (g) *(Note 7)* (phot.mag;em.opt.B) |

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|---|---|---|---|---|---|
| ☑ | ○ | rmag | | mag | (n)(i) [3.8/40.4] Model magnitude in r filter AB scale (r) *(Note 7)* (phot.mag;em.opt.R) |
| ☑ | ○ | imag | | mag | (n)(i) [2.6/39.5] Model magnitude in i filter AB scale (i) *(Note 7)* (phot.mag;em.opt.I) |
| ☑ | ○ | zmag | | mag | (n)(i) [3.5/37.2] Model magnitude in z filter AB scale (z) *(Note 7)* (phot.mag;em.opt.I) |
| ☑ | ○ | e_umag | | mag | (n) [8.9e-8/3444] Mean error on umag (err_u) (stat.error;phot.mag) |
| ☑ | ○ | e_gmag | | mag | (n) [3.8e-8/5955] Mean error on gmag (err_g) (stat.error;phot.mag) |

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| ☑ | ○ | e_rmag | | mag | (n) [5.2e-8/915] Mean error on rmag (err_r) (stat.error;phot.mag) |
| ☑ | ○ | e_imag | | mag | (n) [0/634] Mean error on imag (err_i) (stat.error;phot.mag) |
| ☑ | ○ | e_zmag | | mag | (n) [9.4e-8/3529] Mean error on zmag (err_z) (stat.error;phot.mag) |
| ☑ | ○ | Status | | | [0/4145] Decimal status (resolveStatus) *(Note 5)* (meta.code) |
| ☑ | ○ | spInst | | (char) | Spectral instrument used ("SDSS" or "BOSS") (spec_instrument) *(Note 6)* (meta.id;instr) |
| ☑ | ○ | spType | | (char) | Source type (spec_sourceType) (6) *(Note 8)* (src.class) |
| ☑ | ○ | zsp | 0.01 .. 0.70 | | (n)(i) [-0.012/7.06] Spectroscopic final redshift (when SpObjID>0) (spec_z) *(Note 6)* (src.redshift) |
| ☑ | ○ | e_zsp | | | (n) [-6/30689] Mean error on zsp (negative for bad fit) (spec_zErr) *(Note 6)* (stat.error) |
| ☑ | ○ | f_zsp | ==0 | | (n) [0/902] Zwarning flag (spec_zWarning) (6) *(Note 9)* (meta.code) |
| ☑ | ○ | spCl | ==GALAXY | (char) | Spectroscopic class: "GALAXY"; "QSO"; "STAR" (spec_class) *(Note 6)* (meta.code.class;pos.parallax.spect) |
| ☑ | ○ | subCl | | (char) | Spectroscopic subclass (spec_subClass) (6) *(Note 10)* (src.class) |
| ☑ | ○ | Vdisp | >70 | km/s | (n) [0/850] Velocity dispersion (spec_velDisp) *(Note 6)* (phys.veloc.dispersion) |
| ☑ | ○ | e_Vdisp | | km/s | (n) [-7/4526] Mean error on Vdisp (spec_velDispErr) *(Note 6)* (stat.error) |

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|---|---|---|---|---|---|
| ☑ | ○ | spS/N | | | (n) [-1927/132] Median signal-to-noise over all good pixels (spec_snMedian) *(Note 6)* (em.wl.central;stat.median) |
| ☑ | ○ | M | | | (n) [0/4] Number of USNO-B objects matched within 1arcsec (pm_match) *(Note 11)* (meta.number) |
| ☑ | ○ | pmRA | | mas/yr | (n) [-6121/5519] Proper motion along Right Ascension (pm_pmRa) *(Note 11)* (pos.pm;pos.eq.ra) |
| ☑ | ○ | pmDE | | mas/yr | (n) [-4846/4437] Proper motion along Declination (pm_pmDec) *(Note 11)* (pos.pm;pos.eq.dec) |
| ☑ | ○ | e_pmRA | | mas/yr | (n) [0/4709] Mean error on pmRA (pm_pmRaErr) *(Note 11)* (stat.error;pos.pm;pos.eq.ra) |
| ☑ | ○ | e_pmDE | | mas/yr | (n) [0/4709] Mean error on pmDE (pm_pmDecErr) *(Note 11)* (stat.error;pos.pm;pos.eq.dec) |
| ☑ | ○ | sigRA | | mas | (n) [0/8662] RMS residual of proper motion fit in RA (pm_sigRa) *(Note 11)* (stat.fit.residual) |
| ☑ | ○ | sigDE | | mas | (n) [0/10625] RMS residual of proper motion fit in Dec (pm_sigDec) *(Note 11)* (stat.fit.residual) |
| ☑ | ○ | N | | | (n) [0/6] Number of detections used in the pm fit including SDSS (pm_nFit) *(Note 11)* (meta.number) |
| ☑ | ○ | g(O) | | | (n) [-214740/70189] Magnitude from USNO-B O plate recalibrated to SDSS g (pm_O) *(Note 11)* (phot.mag;em.opt.B) |
| ☑ | ○ | r(E) | | | (n) Magnitude from E plate recalibrated to SDSS r (pm_E) *(Note 11)* (phot.mag;em.opt.R) |
| ☑ | ○ | g(J) | | | (n) Magnitude from J plate recalibrated to SDSS g (pm_J) *(Note 11)* (phot.mag;em.opt.B) |

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|---|---|---|---|---|
| ☑ | ○ | r(F) | | (n) Magnitude from F plate recalibrated to SDSS r (pm_F) *(Note 11)* (phot.mag;em.opt.R) |
| ☑ ALL cols | Reset All | Clear | | (n) indicates a possible blank or NULL column (i) indexed column Submit |
| ☑ | ○ | i(N) | | (n) Magnitude from N plate recalibrated to SDSS i (pm_N) *(Note 11)* (phot.mag;em.opt.I) |
| ☑ | ○ | zph | | (n)(i) [5e-4/1] Photometric redshift estimated by robust fit to nearest neighbors in a reference set (photoz_z) *(Note 12)* (src.redshift) |
| ☑ | ○ | e_zph | | (n) [0.003/0.3] Estimated error of the photometric redshift (photoz_zErr) *(Note 12)* (stat.error) |
| ☑ | ○ | <zph> | | (n) [0.009/0.9] Average redshift of the nearest neighbors; if significantly different from zph this might be a better estimate than zph (photoz_nnAvgZ) *(Note 12)* (src.redshift.phot) |
| ☑ | ○ | chi2 | | (n) [2e-12/6.3e7] Chi-square value for the minimum chi-square template fit (photoz_chisq) *(Note 12)* (stat.value) |
| ☑ | ○ | Q | | [1/3] Quality of the observation: 1=bad; 2=barely acceptable; 3=good (field_quality) *(Note 13)* (meta.code.qual;obs.param;obs) |
| ☑ | ○ | uMJD | d | Date of observation in u band (field_mjd_u) *(Note 13)* (time.epoch) |
| ☑ | ○ | gMJD | d | Date of observation in g band (field_mjd_g) *(Note 13)* (time.epoch) |
| ☑ | ○ | rMJD | d | Date of observation in r band (field_mjd_r) *(Note 13)* (time.epoch) |
| ☑ | ○ | iMJD | d | Date of observation in i band (field_mjd_i) *(Note 13)* (time.epoch) |
| ☑ | ○ | zMJD | d | Date of observation in z band (field_mjd_z) *(Note 13)* (time.epoch) |
| ☑ | ○ | SDSS16 | (char) | (i) SDSS-DR16 name based on J2000 position; provided by CDS (meta.id) |
| ☑ | ○ | SDSS-ID | (char) | (i) SDSS object identifier (run-rerun-camcol-field-object) *(Note 14)* (meta.id) |
| ☑ | ○ | Sp-ID | (char) | (i) Spectroscopic Plate-MJD-Fiber identifier (plate-mjd-fiberID) (**Tip**: to select SDSS spectroscopic sources, enter the condition * `[1-9]*`) (meta.id) |
| ☑ | ○ | MJD | d | Mean observation date (average calculated by CDS from the uMJD, gMJD, rMJD, iMJD, zMJD) (time.epoch) |
| ☑ ALL cols | Reset All | Clear | | (n) indicates a possible blank or NULL column (i) indexed column Submit |

**Adapt form** Display your selection only **Modify** Reset to default columns **Modify**

Display ☑ UCD1+ ☐ UCD1

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→ Rules of usage of VizieR data

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